

derivative in [Figure 3-2](#): it's just a way to measure how quickly a function is growing or shrinking.

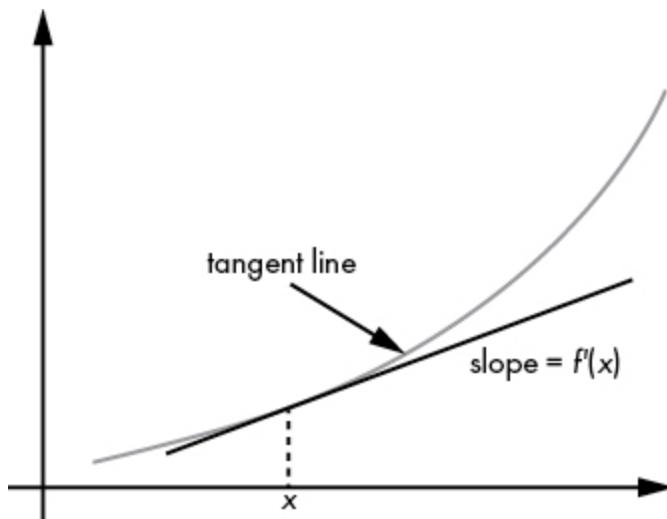


Figure 3-2: To calculate a derivative, we take a tangent line to a curve at a point and find its slope.

We can create a function in Python that specifies this derivative as follows:

```
def revenue_derivative(tax):  
    return(100 * (1/(tax + 1) - 2 * (tax - 0.2)))
```

We used four rules of calculus to derive that function. First, we used the rule that the derivative of $\log(x)$ is $1/x$. That's why the derivative of $\log(\text{tax} + 1)$ is $1/(\text{tax} + 1)$. Another rule is that the derivative of x^2 is $2x$. That's why the derivative of $(\text{tax} - 0.2)^2$ is $2(\text{tax} - 0.2)$. Two more rules are that the derivative of a constant number is always 0, and the derivative of $100f(x)$ is 100 times the derivative of $f(x)$. If you combine all these rules, you'll find that our tax-revenue function, $100(\log(\text{tax} + 1) - (\text{tax} - 0.2)^2 + 0.04)$, has a derivative equal to the following, as described in the Python function:

$$100\left(\left(\frac{1}{\text{tax} + 1}\right) - 2(\text{tax} - 0.2)\right)$$

We can check that the derivative is indeed negative at the country's current taxation rate: